CEF440: Internet & Mobile Programming — 2024/2025

Crowdsensed Drive Test Application

Stakeholder Interview — MNO Network Engineer

Stakeholder: **Mobile Network Operator Engineer**

To understand network monitoring workflows, KPI standards, and data needs of mobile operators.

University of Buea · Dr. Valery Nkemeni · April 11, 2026

Informed Consent & Confidentiality Notice
This questionnaire is part of an academic research project at the University of Buea (CEF440). Your participation is entirely voluntary. All responses will be used solely for academic purposes and will be kept strictly confidential. No personally identifiable information will be published. By completing this questionnaire, you consent to its use in this study. Thank you for your time.

Respondent Information

Name (optional):	
Organisation / Operator:	
Role / Job Title:	
Years of Experience:	
Date:	

Section A: Current Network Monitoring Practices

Q1.

What methods does your organisation currently use to assess mobile network coverage? (Select all that apply)

- ■ Physical Drive Testing
- ■ Walk Testing
- ■ Network Management System (NMS) Monitoring
- ■ Customer Complaint Analysis
- ■ Crowdsourced Data (e.g. OpenSignal)
- ■ Other (specify below)

Q2.

How frequently are drive tests conducted in your network area?

- ■ Daily
- ■ Weekly
- ■ Monthly
- ■ Quarterly
- ■ Only when issues are reported

Q3.

What are the most significant limitations of your current drive testing approach?

Q4.

Which geographic areas in your coverage zone are most difficult to assess? Why?

Section B: Key Performance Indicators (KPIs)

Q5.

Which of the following network parameters are most critical for your coverage assessments? (Rank 1=most important)

- ■ RSRP (Reference Signal Received Power)
- ■ RSRQ (Reference Signal Received Quality)
- ■ SINR (Signal-to-Interference-plus-Noise Ratio)
- ■ RSSI (Received Signal Strength Indicator)
- ■ Throughput (Download/Upload Speed)
- ■ Cell ID / eNodeB ID
- ■ Timing Advance (TA)
- ■ Handover success rate

Q6.

What RSRP threshold do you use to classify an area as a 'coverage hole'?

- ■ Below -100 dBm
- ■ Below -105 dBm
- ■ Below -110 dBm
- ■ Below -115 dBm
- ■ Below -120 dBm
- ■ Other (specify)

Q7.

What minimum sampling rate (measurements per unit distance/time) would make crowdsourced data useful to you?

Q8.

What data format do you prefer for receiving coverage reports? (e.g., CSV, JSON, GeoJSON, shapefile, dashboard)

Section C: Crowdsensing Acceptance

Q9.

How willing would your organisation be to use crowdsourced smartphone data to supplement drive tests?

Not willing at all	■ 1	■ 2	■ 3	■ 4	■ 5	*Extremely willing*

Q10.

What accuracy level of GPS geo-referencing would you require for crowdsourced measurements to be acceptable?

- ■ Within 1 metre
- ■ Within 5 metres
- ■ Within 10 metres
- ■ Within 50 metres
- ■ Accuracy not critical

Q11.

What are your primary concerns about relying on crowdsourced data? (Select all that apply)

- ■ Data accuracy / calibration
- ■ Device variability across manufacturers
- ■ Data privacy and user consent
- ■ Inconsistent geographic coverage
- ■ Lack of standardisation
- ■ Security of data transmission
- ■ None — no concerns

Q12.

What conditions would need to be met for your organisation to officially adopt crowdsensed data in network planning?

Section D: Data Management & Integration

Q13.

Does your organisation currently use any crowdsourced data providers (e.g., OpenSignal, Ookla, GSMA)? If yes, which?

Q14.

Would you be willing to integrate a local university-developed crowdsensing platform into your monitoring workflow for a trial period?

- ■ Yes, definitely
- ■ Yes, possibly
- ■ No, unlikely
- ■ No, definitely not

Q15.

What backend data visualisation features would be most valuable to your team? (e.g., heatmaps, coverage hole alerts, time-series trends)

Q16.

Are there any regulatory or data protection requirements we should be aware of when collecting network data from users in Cameroon?

Section E: Additional Insights

Q17.

From your professional experience, what single feature would make a crowdsensed drive test application most valuable to network operators?

Q18.

Is there anything else you would like to share that could help guide the development of this application?

Thank you for your time and valuable input. Your responses will directly inform the design of this application.